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WRF Cold-Pool Sensitivities Relevant to Winter Ozone Modelling in the Uinta Basin

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Abstract

Cold-pool structure controls the confinement of wintertime ozone and its precursors in Utah's Uinta Basin. Previous studies have documented difficulty capturing stable cold pools in meteorological models, complicating air quality simulations when pollutants disperse too readily. Herein, we evaluate the meteorology behind one such high-ozone event on 2 February 2013 with Weather Research and Forecasting (WRF) simulations; the chemistry is not simulated. We test the sensitivity to Global Forecast System (GFS) or North American Mesoscale (NAM) forcing, domain feedback, boundary-layer and surface-layer schemes, slope-aware radiation, and terrain datasets (fine versus coarse). Across five hourly samples, GFS-driven runs were warmer at the evaluated surface sites, had an approximately 3.6 K smaller 50–500 m potential-temperature endpoint difference, and had a 0.8–1.0 m s^{-1} larger layer-mean wind-vector magnitude than feedback-matched NAM runs. The GFS also retained a lower mean heat deficit, indicating a weaker modelled cold pool. The heat deficit increased in all four runs from 1400 to 1800 UTC; hence the product separation suggests a state difference without demonstrated faster cold-pool erosion. Toggling feedback and parametrisation settings yielded smaller responses; terrain fineness produced spatial responses that station comparisons could miss. The results guide diagnostics for subsequent air-chemistry modelling, while leaving initial versus lateral forcing, cold-pool depth, and basin export for future studies.

Keywords: winter ozone; WRF simulation; meteorological uncertainty; cold-air pool; complex terrain; model resolution

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1. Introduction

Wintertime ozone in the Uinta Basin develops within snow-covered, stable lower-atmospheric layers that confine precursors near the surface [1–3]. Here we study the meteorology of one such event; the chemistry is not simulated. The basin floor, benches, and surrounding terrain support strong temperature contrasts and thermally driven slope flows [4,5]. Comparable winter ozone episodes have been documented in Wyoming's oil-and-gas-producing basins [6,7]; late-winter ozone pollution has also occurred in northern China under different chemical conditions [8]. Herein, we examine a well documented 2013 Uinta Basin episode whose meteorology challenged earlier simulations [2,9,10]. The five hourly samples span early morning through late morning, when daytime heating can begin to erode the cold pool.

Surface observations alone cannot estimate cold-pool depth or the three-dimensional mixing volume supplied to a chemistry model. Evaluation therefore requires complementary diagnostics of potential temperature, lower-layer winds, and heat storage, with a clear distinction between observed agreement and modelled structure.

Earlier Uinta Basin ozone modelling demonstrates strong case dependence, so individual cases cannot establish general configuration recommendations [2,9,10]. Related work on particulate matter in Salt Lake City [11] and national air quality prediction systems [12,13] places these meteorological questions within broader efforts to improve coupled models. Snow cover can strengthen cold pools by increasing albedo, while cloud phase modulates longwave forcing and nocturnal cooling [2]. Snow also affects the light available for ozone production, whose chemical sensitivity changes through the winter [14]. These coupled processes motivate joint meteorological and chemical observations in western basins [15]. Further compounding potential error, the choice of microphysics and domain configuration alters both the cold pool and its representation on atmospheric-chemistry grids. Four-dimensional data assimilation (FDDA) depends on the nudged variables, input analysis, and scheme-dependent boundary-layer diagnosis [10]. A subsequent study examined reinitialisation, snow initialisation, and nudging, but assimilated evaluation data and differing samples limit configuration comparisons [16]. These studies motivate the present sensitivity tests: *which model choices most affect the meteorology supplied to a simulation of a cold-pool and high-ozone event*?

Errors in meteorology, emission inventory, photolysis, etc., may cancel in coupled simulations. While these errors are difficult to extricate, a common difficulty is representing persistent cold-air pools [2,10]. Changing emissions with fixed meteorology can improve ozone agreement without uniquely identifying inventory error [1]. Cumulative changes to these processes complicate attribution, and improved temperature profiles need not improve ozone profiles [9]. Hence credible meteorology reduces this confounding but does not guarantee improved ozone forecasts.

Deploying the Weather Research and Forecasting (WRF) model [17], we test whether the lower-atmospheric response to two different sets of initial and lateral-boundary conditions recurs under one-way (i.e., top-down) and two-way nesting, and compare it with parametrisation-suite and terrain-fineness responses. The diagnostics combine surface skill, the 50–500 m potential-temperature difference, the layer-mean horizontal wind-vector magnitude, and the mean heat deficit over a delineated area. Together, they describe the simulated cold-pool state, but in focusing our scope, we do not estimate ventilation or basin export herein.

We ask:

1. Does the lower-layer response to different weather-model forcing datasets recur under both feedback modes?
2. How do changes in domain feedback, boundary-/surface-layer schemes, and slope-aware parametrisations compare with the response to different weather-model forcings?
3. How does changing the terrain dataset fineness affect the same scenario?
4. What do the combined diagnostics reveal that in situ stations may not?

The study contains no chemistry calculation and does not separate initial from lateral-boundary forcing; follow-on work in preparation will connect meteorological representation with air-chemistry modelling for this and related cases.

2. Materials and Methods

2.1. Case, Observations, and Evaluation Domains

The case falls within the January–February 2013 period of snow cover, cold pools, and elevated Uinta ozone [1,2,10]. Integrations capture 1200–1800 Coordinated Universal Time (UTC) or 0500–1100 Mountain Standard Time (MST; UTC−7) on 2 February 2013. Excluding the initialised output and first-hour forecast to reduce spin-up influence leaves 1400–1800 UTC (0700–1100 MST), the longest common retained window (i.e., five hourly samples over four elapsed hours). This morning window samples the transition towards daytime heating but cannot establish a diurnal cycle of cold-pool development and breakup. Data from in situ stations acquired from Synoptic Data include the 2 m temperature at 11 sites, 10 m wind speed at 10 sites, and surface pressure at 9 sites [18]. The temperature sample includes every registered station reporting in the window; Sand Wash had no reports. Subhourly reports were averaged within each UTC clock hour. The stations span the basin floor, sidewalls, and benches but are unevenly distributed; their skill scores describe this sample rather than the basin as a whole. We use the 1 km d02 domain because seven stations lie outside the usable d03 interior (Figure 1). Four stations provide temperature context from 31 January 0000 UTC to 4 February 0000 UTC (Figure 3). For the observed and modelled Mountain Home-minus-Pariette Draw index (Figure 6), we calculate the surface potential temperature as $\theta = T(1000/p)^{0.2854}$, with T in kelvin and p in hPa, using the dry-air Poisson relation [19].

The ozone context uses the US Environmental Protection Agency (EPA) Air Quality System (AQS) AirData daily archive for January–February 2013 [20]. We retained the "Ozone 8-h 2015" daily-summary row at Ouray, Redwash, Roosevelt, Vernal and Little Mountain: one row per site on each of 59 local dates, with a within-day coverage of 47–100% marked in the figure. The satellite panel below (Figure 2) uses the National Aeronautics and Space Administration (NASA) Global Imagery Browse Services (GIBS) Terra Moderate Resolution Imaging Spectroradiometer (MODIS) corrected-reflectance 7-2-1 layer for 2 February 2013 [21]. We use spectral absorption and scattering principles described by Liou [22], with snow/cloud colours specified by NASA's layer documentation [23] to inform interpretation.

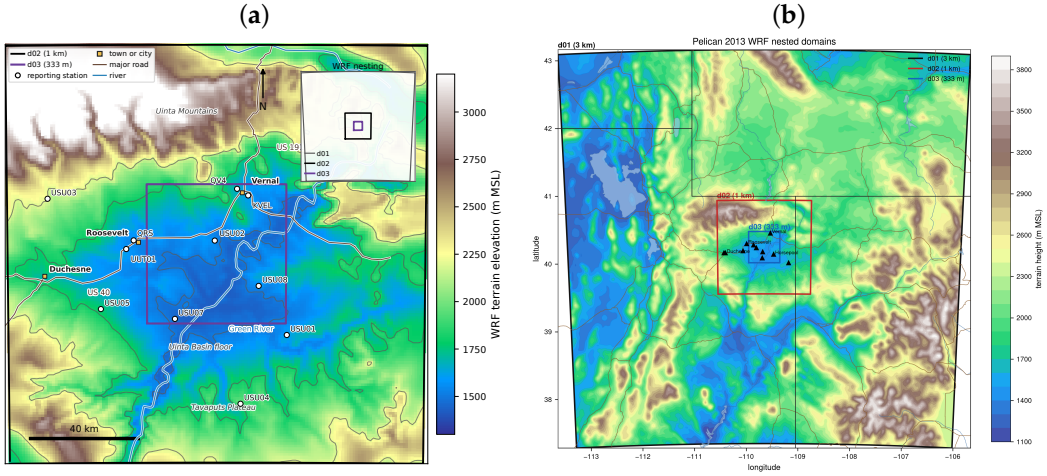


Figure 1. Geographic and sampling orientation for the evaluation. (**a**) The d02 terrain, reporting stations, and d03 interior define the common evaluation setting. (**b**) The d01–d02–d03 WRF grid layout places that setting within the full three-domain configuration. The d02 domain contains all eleven reporting stations; d03 contains four usable station columns.

2.2. WRF Configuration and Archived Simulation Family

We used WRF version 4.8.0 [17], initialised at 1200 UTC with lateral-boundary conditions through 1800 UTC on 2 February 2013. The inputs were the 1200 and 1800 UTC zero-hour GFS grid 004 (0.5°; `gfsanl_4`) or NAM grid 218 (12 km; `namanl_218`) analyses [24,25]. These integrations include temporally interpolated lateral-boundary forcing, so the retained response combines initial-state and boundary influences. Three nested domains, d01–d03, have horizontal spacings of 3 km, 1 km, and 333.3 m, with 75 model (eta) levels. Note that GFS and NAM inputs supply 27 and 40 metgrid levels, respectively. The WRF grid has approximately 20 m vertical spacing near the surface and a prescribed 50 hPa model top. The control physics-parametrisation suite comprises Thompson microphysics [26], Rapid Radiative Transfer Model for General Circulation Models (RRTMG) longwave and shortwave radiation [27], the revised fifth-generation Pennsylvania State University–National Center for Atmospheric Research Mesoscale Model (MM5) similarity surface layer [28], the Noah land-surface model [29], and the Yonsei University (YSU) planetary boundary layer (PBL) scheme [30]. No nudging or data assimilation was used. Table 1 lists the simulations analysed here.

Table 1. Simulations analysed in this study. Every run uses WRF 4.8.0, three nests at 3/1/0.333 km with 75 levels, Thompson microphysics, RRTMG radiation, and Noah land surface unless stated. "Fine terrain" is the custom 3-arc-second source, "default terrain" is the 30-arc-second class, and "coarse terrain" is the 5-arc-minute GMTED2010 source.

Code	Product, Feedback	Terrain and Physics
X0	NAM, one-way	fine terrain, YSU/MM5
X1	NAM, one-way	X0 plus slope radiation and shading
X2	NAM, one-way	X1 with MYJ/Eta instead of YSU/MM5
X3	NAM, one-way	default-terrain reference configuration
X4	NAM, one-way	deliberately coarse 5-arc-minute terrain
X5	NAM, two-way	default terrain
X6	GFS, two-way	default terrain
X9	NAM, two-way	X0 with two-way feedback
X10	GFS, one-way	default terrain

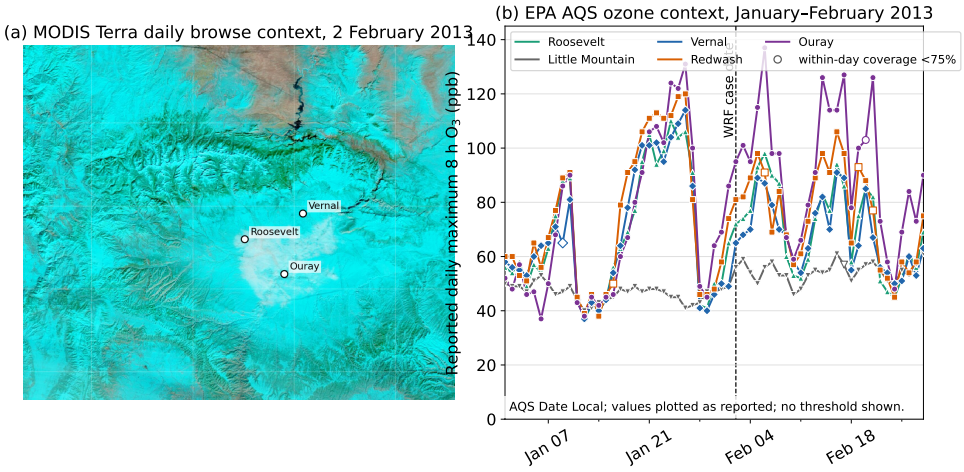


Figure 2. Observed context for the 2 February 2013 simulation date. (**a**) NASA GIBS Terra MODIS corrected reflectance, bands 7-2-1, for the 2 February calendar day; snow and ice generally appear cyan, most liquid cloud appears white, and ice cloud may also appear cyan. (**b**) EPA AQS first maximum value (the eight-hour ozone daily-summary row) at Ouray, Redwash, Roosevelt, Vernal, and Little Mountain. These values use the current EPA daily-summary archive rather than a reconstruction of earlier studies' observation processing. Hollow markers identify the seven site-days with less than 75% within-day observation coverage.

The terrain datasets of varying fineness were ingested by the WRF Preprocessing System. The fine, default, and coarse sources have spacings of 3 arc seconds, a 30-arc-second class, and 5 arc minutes, respectively; the coarse source is GMTED2010 [31]. These correspond to approximately 90 m, 900 m, and 9 km north–south spacing at 40°N, distinct from WRF horizontal-grid spacing.

Hence, our primary comparison crosses GFS and NAM forcing with one-way and two-way feedback on default terrain (X3, X5, X6, and X10). A controlled contrast pairs runs differing in one designated treatment on the same domain, locations, hours, and diagnostic calculation. Snow and other land-surface initial fields remain bundled with the forcing product; the Noah scheme is held fixed.

2.3. Surface and Common-AGL Diagnostics

Surface variables and vertical fields were sampled from the same nearest d02 ($\Delta x = 1$ km horizontal spacing) column at each station, keeping the model terrain and profile consistent. The station-to-cell distances are 133–594 m, with default-terrain elevation offsets of −32.4 to +14.3 m (Appendix A Table A6). These pairings retain differences in exposure, slope, and land cover; alternative interpolation methods were not tested. The potential temperature was interpolated at the layer bounds; wind components were linearly interpolated every 25 m from 50 to 500 m above ground level (AGL) for height integration. We repeat the eleven-column, five-hour comparison for all nine combinations of 50–500, 50–250, and 100–500 m layers with 10, 25, and 50 m wind integration increments. The bounds define a common shallow layer rather than an observed inversion boundary.

The finite-layer potential-temperature difference is defined as

$$S_{50:500} = \theta(500\ \mathrm{m}) - \theta(50\ \mathrm{m}). \tag{1}$$

An increase in potential temperature with height indicates dry static stability [19,32]. The endpoint difference does not locate an inversion top or resolve internal structure: a strong shallow inversion can share the same endpoints as a different profile. In addition, the magnitude of the layer-mean horizontal wind vector was defined as

$$U_{50:500} = \left(\bar{u}^2 + \bar{v}^2\right)^{1/2}, \tag{2}$$

where the bars denote the height means from 50 m to 500 m AGL, not temporal or turbulent averages. We also calculate the layer-mean scalar speed $\overline{(u^2 + v^2)^{1/2}}$ and the vector-to-scalar ratio to assess cancellation between winds at different heights. Native PBL height serves as a scheme-dependent companion diagnostic rather than an independent inversion-depth estimate [10].

2.4. Pairing, Aggregation, and Short-Sample Sensitivity

Run differences were paired by location and valid hour, and then reduced to one spatial median per hour. The reported five-hour effects are means of those medians; stations are not spatially independent. To explore the sensitivity to temporal dependence [33], a circular moving-block calculation drew 20,000 resamples of the five medians using two-hour blocks. Its 2.5th and 97.5th percentiles are descriptive sensitivity bounds for this single case; we emphasise the practical effect size without treating them as evidence of statistical significance.

2.5. Area-Integrated Heat Deficit

To represent the cold-pool strength numerically, we adapted the basin heat-deficit formulation of Whiteman et al. [34] and the column implementation of Adler et al. [35].

Using the hydrostatic balance, $dp = -\rho g\, dz$, to express the column mass in pressure coordinates [19], we defined

$$H = \frac{c_p}{g} \int_{p_c}^{p_s} \max\{\theta_{\text{crest}} - \theta(p), 0\}\, dp, \tag{3}$$

where c_p is the dry-air specific heat at constant pressure, g is the gravitational acceleration, and $p_s > p_c$ are the surface and crest-reference pressures. The reference potential temperature is interpolated to the fixed crest height in each modelled column; clipping retains only positive deficits. This potential-temperature proxy then describes stratification relative to that reference. We report its cell-area-weighted mean in MJ m^{-2}: the primary area is the convex hull of eleven stations and eleven basin waypoints, restricted to terrain below 2200 m (4965 d02 cells covering 5277 km^2). We tested the sensitivity of the results to this hull at 2100, 2200, and 2300 m; increasing or eroding the area by one cell; and in station-only areas at 2200 m. Given the sensitivity of the case to complex terrain, we ensure that the GFS–NAM contrasts share a terrain dataset.

3. Results

3.1. Observed Episode Context

The MODIS image shows extensive snow and ice across the basin and surrounding terrain (Figure 2a). The AQS series places 2 February within a broader period of elevated ozone, with daily maximum eight-hour concentrations of 56–95 ppb across the five sites (Figure 2b). These observations establish episode context without attributing ozone to the simulated meteorology.

3.2. Observed Surface-Temperature Context

Inspecting a sample of observation stations across a range of altitudes, we found that a representative basin-floor station repeatedly separated from the higher sites between 31 January and 3 February, with a strong variation in diurnal temperature differences (Figure 3). Seep Ridge (USU04; 1933 m) appears anomalous, despite a modest station-to-cell distance of 133 m and 3 m terrain offset in the vertical, showing latent limitations in estimating cold-pool strength with observations. Exposure, slope flow, and representativeness are possible explanations, but the metadata do not distinguish them. A fresh provider query returned all 17 retained-window samples per requested variable without QC flags, but supplied no exposure history or independent accuracy check. The recurring floor-to-bench separation is consistent with a cold-pool surface pattern; however, the horizontally separated sites cannot determine its vertical structure.

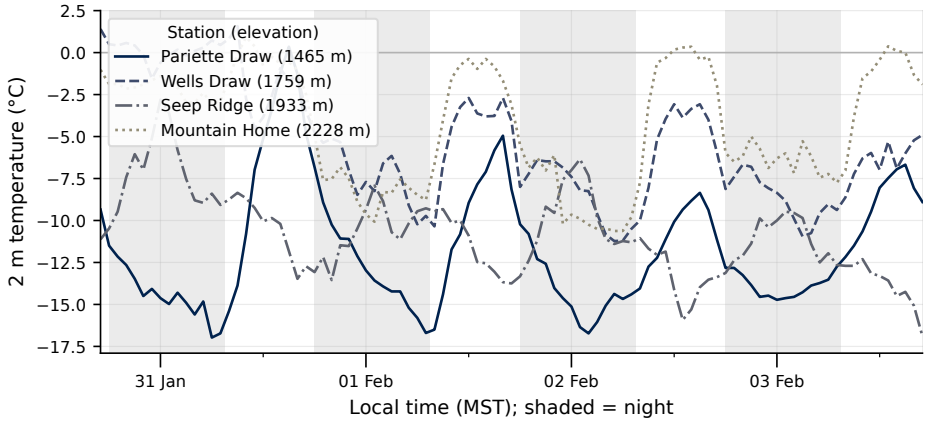


Figure 3. Observed 2 m temperature evolution at four elevation-spanning Uinta Basin stations from 31 January through 3 February 2013. Shading indicates approximate local night. The series show a spatial surface-temperature pattern, not a co-located vertical profile.

3.3. Overview of Experiment Responses

The forcing-product contrasts give the largest temperature and layer-mean wind responses (Figure 4). The PBL/surface-layer response is smaller; slope-radiation and feedback interaction effects cluster near zero on these scales. Pressure and wind-error metrics have inconsistent trends in time, preventing an overall configuration ranking for this case.

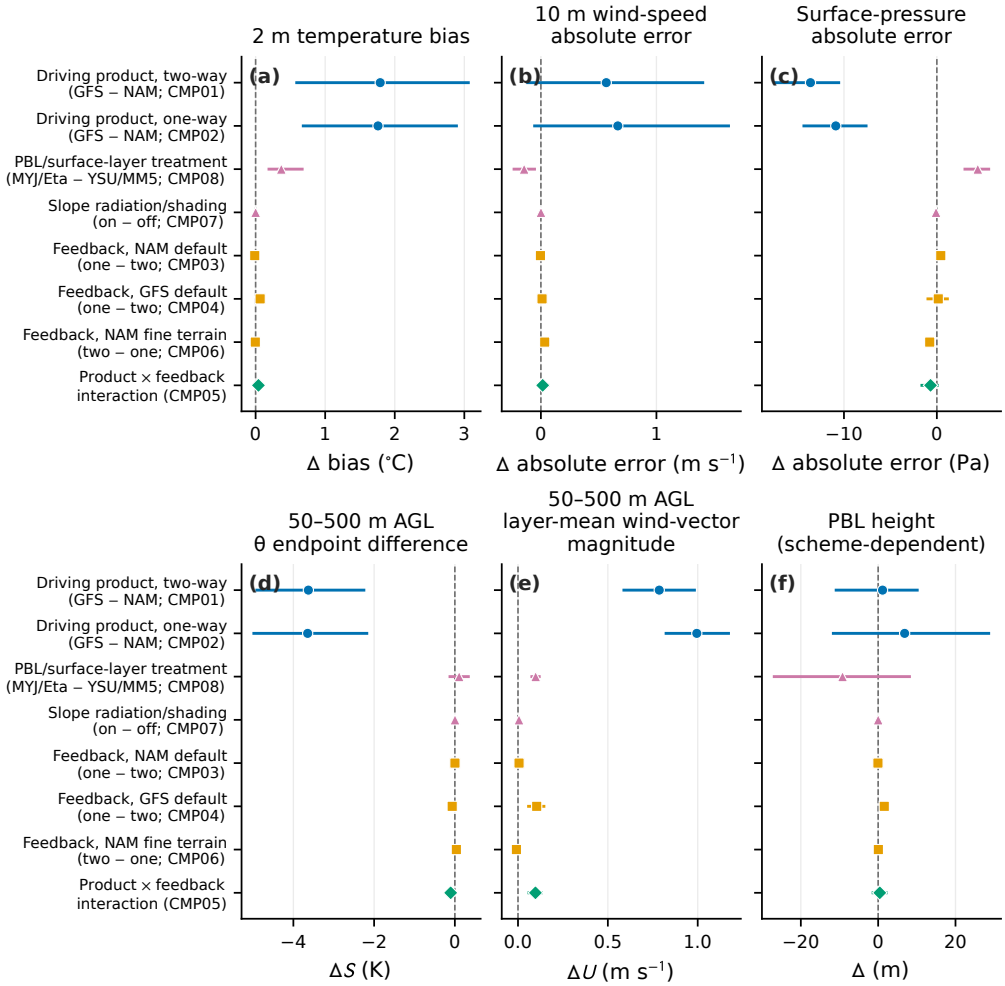


Figure 4. Mean five-hour effects for the controlled comparisons that share the station/common-AGL pipeline. Colours and marker shapes identify forcing-product, feedback, and interaction contrasts. Whiskers show descriptive block-sensitivity intervals for five dependent hourly medians.

3.4. Driving-Product Response in the Lower Layer

Runs forced with GFS produced a smaller 50–500 m potential-temperature difference than NAM under both feedback modes (Figure 5). Paired GFS-minus-NAM effects averaged −3.63 K with two-way feedback and −3.65 K with one-way feedback, consistently borne out by a constant sign at every hour (Table A1). The layer-mean wind-vector magnitude increased by 0.79 and 0.99 m s^{-1}, respectively, also with consistent hourly signs. Scalar-speed differences were similar, and directional cancellation therefore likely does not create the speed ordering (Appendix A Table A2). The vector-to-scalar ratio difference was not consistent and hence not analysed further.

The paired GFS-minus-NAM endpoint and wind differences retain their respective signs in every hour across all nine layer/increment combinations. Their magnitudes depend on layer bounds (Appendix A Table A3); varying the integration increment at fixed bounds changed the mean paired wind effects by less than 0.02 m s^{-1}. This tests the integration sampling of the same linearly interpolated winds, rather than the native vertical resolution or an alternative interpolation method. Native PBL-height differences change sign under both feedback modes, consistent with the diagnostic's scheme dependence discussed by Tran et al. [10].

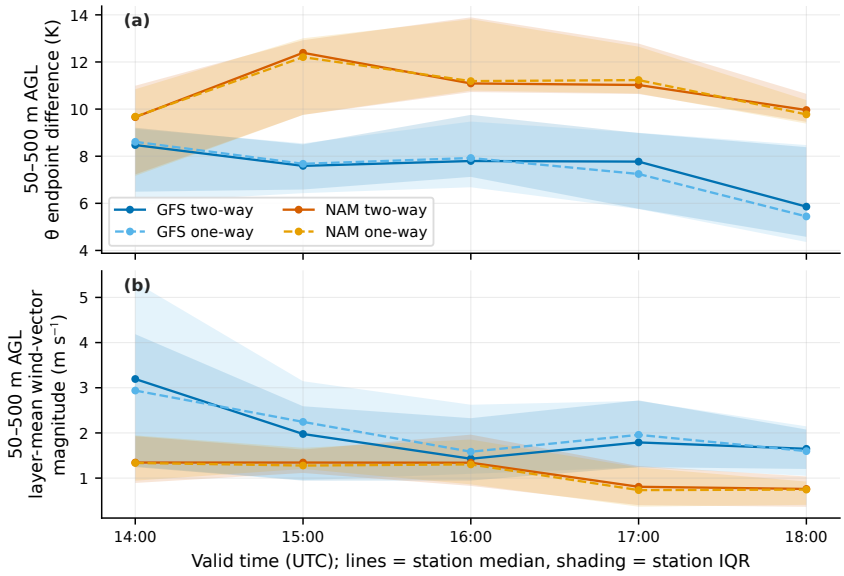


Figure 5. Station-column medians and interquartile ranges for the 50–500 m AGL potential-temperature endpoint difference and layer-mean wind-vector magnitude over the matched window. Solid lines denote two-way feedback and dashed lines denote one-way feedback. Both feedback modes retain the GFS–NAM ordering throughout the window.

3.5. Surface and Spatial Corroboration

Runs forced with GFS were warmer at the surface than NAM, consistent with the common-layer diagnostics (Table A4). The 2 m temperature-bias contrasts were approximately +1.8 °C under both feedback modes. The pressure's absolute error decreased, while the hourly wind-error differences followed no strong pattern. Figure 6 compares the observed bench-minus-floor surface index with the two-way-feedback simulations.

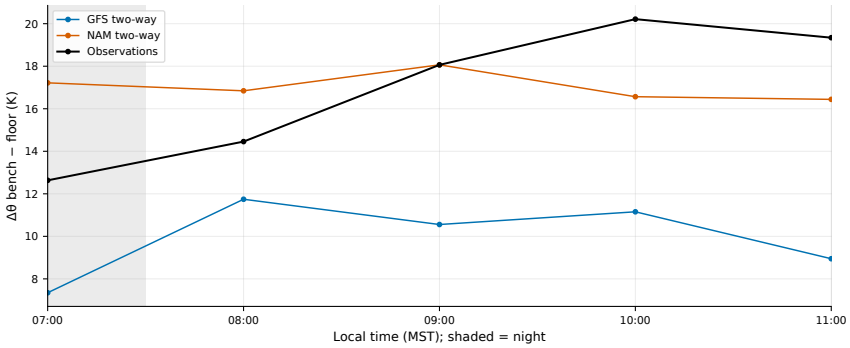


Figure 6. Observed and modelled Mountain Home-minus-Pariette Draw surface potential-temperature contrast from 1400 to 1800 UTC on 2 February 2013. The modelled series are X6 (GFS two-way) and X5 (NAM two-way) on d02, sampled at the same five hours as the observations. The index is a horizontally separated bench-minus-floor surface contrast.

The five-hour mean bench-minus-floor contrast was 16.94 K observed, 17.03 K in X5 (NAM two-way), and 9.95 K in X6 (GFS two-way). Across the area of interest, GFS-minus-NAM near-surface responses were positive on average and concentrated below 500 m AGL under both feedback modes (Table A4).

3.6. Heat Deficit

Runs driven by GFS retained a lower area-mean heat deficit at every hour in all sensitivity tests (i.e., changing the area's footprint), with mean paired effects of approximately

−1.2 MJ m^{-2} under both feedback modes (Table A5). Every run ended with a greater deficit at 1800 than at 1400 UTC, despite intermittent within-window declines (Figure 7).

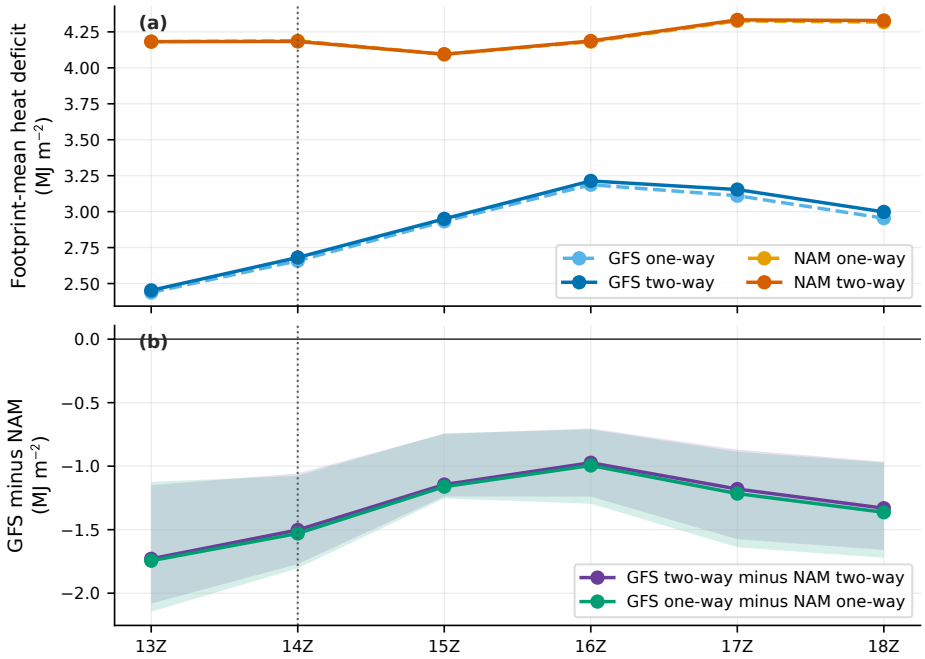


Figure 7. Footprint-mean heat-deficit trajectories for the four default-terrain matrix members and seven-definition sensitivity envelopes for the two GFS-minus-NAM pairs. The 1300 UTC values orientate the trajectories but are excluded from every reported criterion and effect. GFS-driven runs retain a smaller deficit throughout, but the deficit does not decline faster in either GFS-based member.

3.7. Fine-Terrain Experiments

Replacing YSU/revised-MM5 with Mellor–Yamada–Janjić (MYJ)/Eta schemes [36] on fine terrain with slope radiation (X2 minus X1) increased the temperature bias by 0.37 °C, reduced the wind absolute error by 0.145 m s^{-1}, and increased the layer-mean wind-vector magnitude by 0.098 m s^{-1}. All changes were consistent in the 5 h period, but were smaller than the forcing-product impact. The inconsistent temperature- and wind-error changes preclude further commentary on the overall scheme behaviour.

Adding slope radiation and shading (X1 minus X0) had negligible effects at basin-floor station columns but produced a spatially organised sidewall skin-temperature response.

3.8. Nest-Feedback and Terrain Fineness

Effects introduced by toggling domain two-way feedback were small for both products: the one-way minus two-way contrasts were at most 0.07 K in bulk potential-temperature difference and 0.10 m s^{-1} in wind-vector magnitude (Table A1). The fine-terrain NAM comparison (X9 minus X0, two-way minus one-way) was similarly small, with block-sensitivity bounds spanning zero for both metrics. The surface wind and pressure changes were negligible, providing no feedback recommendation beyond this case.

3.9. Terrain-Source Tests

The terrain comparisons hold NAM forcing, one-way feedback, WRF grid spacing, and YSU/MM5 physics fixed as control. A default-minus-fine (X3 minus X0) experiment had a small d02 response, while a wider default-minus-coarse (X3 minus X4) and fine-minus-coarse (X0 minus X4) terrain experiment gave larger spatial responses. For fine minus coarse at 1600 UTC, the heat-deficit median changed sign between d02 (+0.014 MJ m^{-2}) and d03 (−0.184 MJ m^{-2}), while the two d03 section medians were +0.278 and +0.369 K. Figure 8

shows this domain-dependent response; it does not establish a drainage mechanism or a benefit from finer terrain.

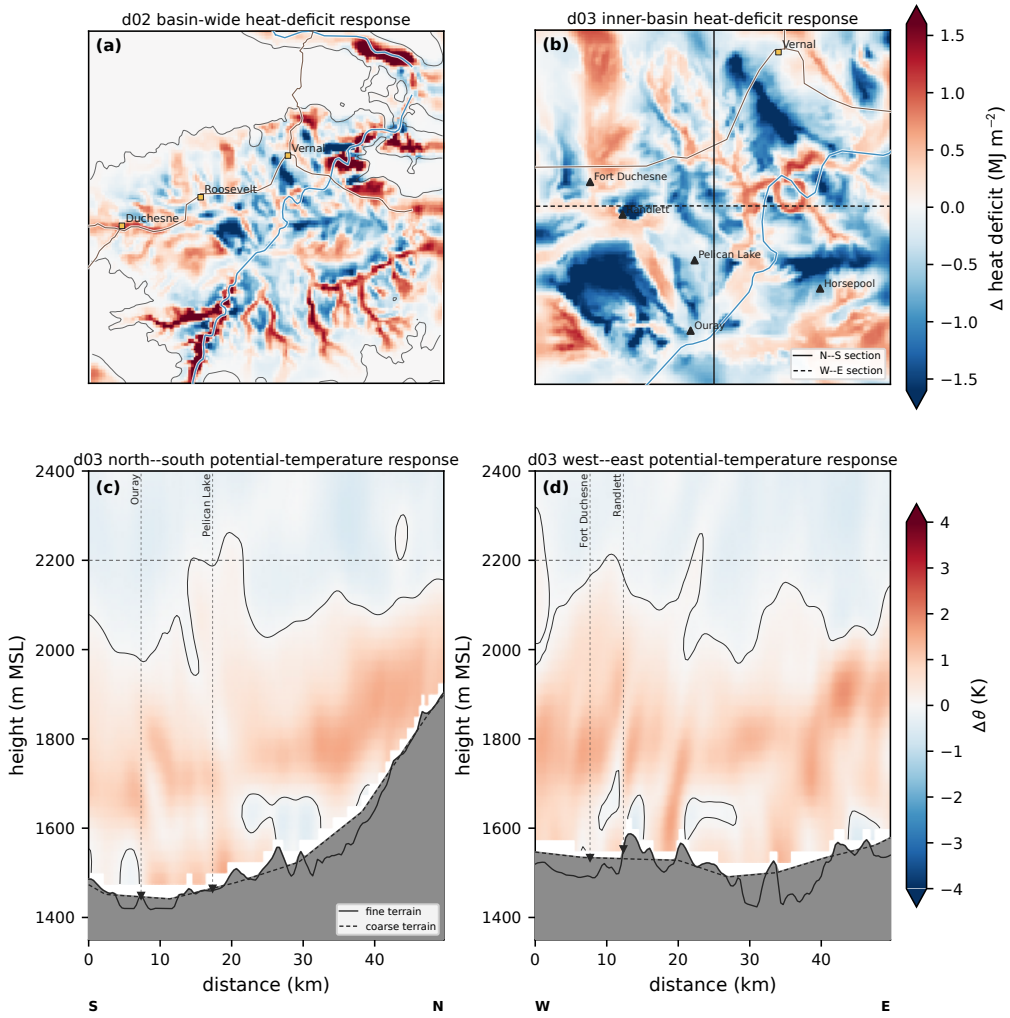


Figure 8. Diagnostic response to the widest controlled terrain comparison at 1600 UTC, calculated as X0 minus X4 (fine 3-arc-second minus coarse 5-arc-minute source). (**a**) The basin-wide d02 and (**b**) inner-basin d03 top-down heat-deficit maps share a fixed ± 1.6 MJ m^{-2} scale. (**c**) The north–south and (**d**) west–east d03 lower-atmosphere potential-temperature sections share a fixed ± 4 K scale and the 2200 m crest reference.

4. Discussion

4.1. Forcing-Product Differences in the Lower Layer

The principal result is the GFS–NAM separation under both feedback settings: the GFS has a smaller potential-temperature endpoint difference, larger layer-mean wind-vector magnitude, and lower heat deficit (weaker modelled cold pool). These differences matter for the meteorology supplied to a chemistry model, but do not measure the precursor confinement or residence time. Because emissions, snow, cloud, and photolysis/chemical-mechanism errors can compensate in coupled simulations [1,2,9], forcing-product sensitivity deserves focus in future air-chemistry simulations; ensemble experiments across multiple cases would test its generality.

The modelled d03 Horsepool profiles, sampled near the lowest point in the Basin (and cold-air sink), show product-specific initial states and first-hour evolution (Figure 9). The early profile differences reinforce the need to assess the meteorological inputs before adding chemistry, but cannot separate initial-state influence from lateral-boundary evolution.

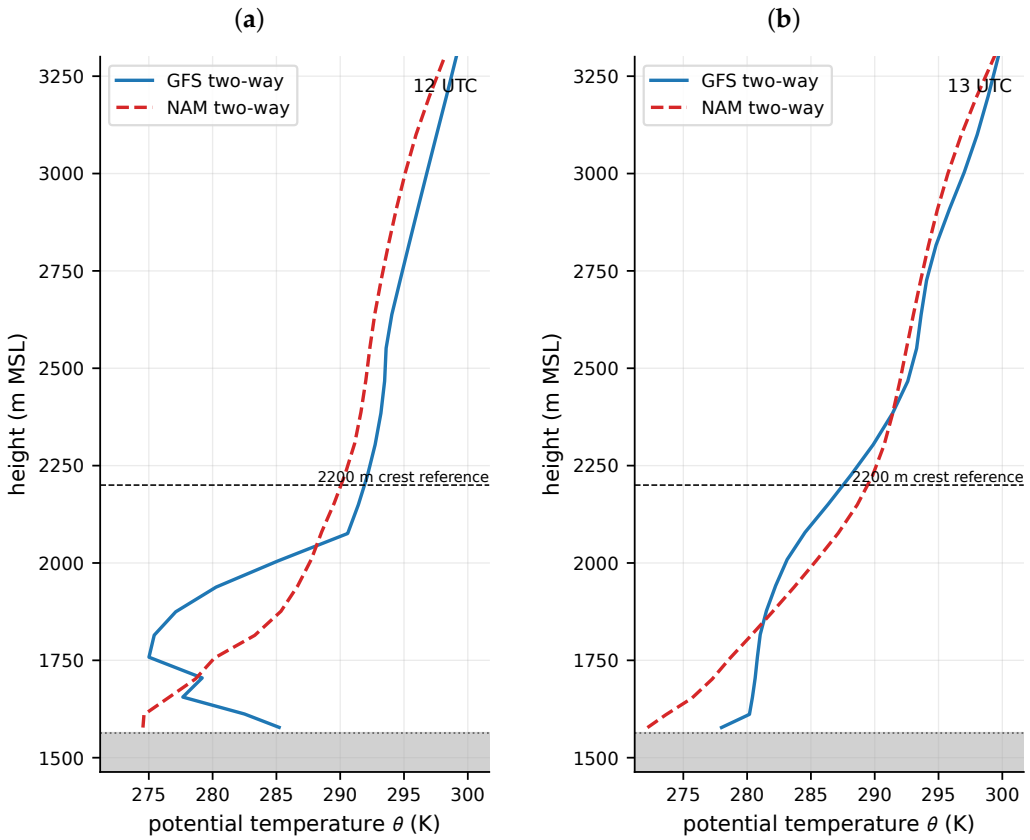


Figure 9. Model d03 ($\Delta x = 333.3$ m) potential-temperature profiles at Horsepool for X6 (GFS) and X5 (NAM). (**a**) The 1200 UTC initialised states and (**b**) the 1300 UTC one-hour forecasts use identical axes. Grey shading marks levels below the local model terrain, and the dashed 2200 m line is the fixed crest reference rather than an observed cold-pool top.

4.2. Heat-Deficit Evolution and Spatial Extent

A lower GFS-run heat deficit persisted across all footprint sensitivity tests, but positive net changes in every run do not demonstrate faster cold-pool erosion during the retained window. Indeed, at least a full diurnal cycle is needed to test differences in cold-pool erosion or persistence, and is a topic of the authors' current research. The forcing-product response was strongest below 500 m AGL and consistent in its mean effect, while its early spatial sign share fell below the predeclared two-thirds criterion (Table A4).

4.3. Parametrisation, Feedback, and Terrain Sensitivities

Parametrisation and feedback effects were smaller than the forcing-product response effect on the temperature and wind diagnostics. The changes in parametrisation configuration modulated wind error while increasing temperature separation; feedback remained small on both terrain branches. While the changes were small, slope radiation and terrain-fineness effects produced spatial responses that in situ stations could miss. These results favour combining station and spatial diagnostics for a fully three-dimensional analysis (and advancing in time). Finer terrain can reveal geometric detail hidden by a coarse representation, as illustrated by scale-dependent measurements of irregular boundaries [37]. That analogy does not establish aerodynamic roughness or drainage speed in WRF; whether finer terrain changes cold-air pathways or storage remains a question for targeted experiments.

4.4. Observational and Design Limits

A bounded search of accessible holdings found no measured in-basin profile within the exact retained window. A provisional Roosevelt sounding began at 1801:54 UTC [38],

just after the window, and has not undergone scientific QC or matched WRF analysis; inaccessible holdings were not treated as empty. Thus the surface comparisons do not validate inversion depth or free-air vertical structure. Separating initial from lateral forcing requires new experiments, while nudging tests require independent observations withheld from assimilation. Longer matched runs and additional cases are needed to assess persistence and generality.

5. Conclusions

We have presented WRF experiments to highlight the sensitivity of cold-pool representation—relevant to the retention of ozone and its precursors—to combined initial and lateral weather-model forcing. We use a well documented winter 2013 case in the Uinta Basin to identify differences among the meteorological simulations. The GFS–NAM contrast produced the largest repeated lower-layer response on the d02 (1 km) domain. Under both feedback modes, GFS-driven runs were warmer at the evaluated surface sites and had an approximately 3.6 K smaller 50–500 m potential-temperature difference, a 0.8–1.0 m s^{-1} larger wind-vector magnitude, and an approximately 1.2 MJ m^{-2} lower area-mean heat deficit (weaker cold pool). The heat-deficit separation represents a state offset without demonstrated faster cold-pool erosion.

Nest-feedback effects were small (at most 0.07 K and 0.10 m s^{-1} on the same layer metrics), while MYJ/Eta increased the temperature bias by 0.37 °C and reduced the wind absolute error by 0.145 m s^{-1}. The slope radiation and terrain effects depended on the spatial definition: the fine-minus-coarse heat-deficit median changed from +0.014 MJ m^{-2} on d02 to −0.184 MJ m^{-2} on d03. These findings support complementary station, layer, and basin-wide diagnostics for meteorology supplied to chemistry models, followed by ensemble experiments across multiple cases to examine uncertainty. Targeted forcing experiments, longer matched integrations, and independent vertical observations are needed to identify forcing contributions and evaluate inversion depth and persistence; this study establishes neither basin export nor ozone response. Appendix A retains numerical summaries and sampling metadata for reproducibility.

Appendix A. Data Tables and Secondary Diagnostics

Table A1. Paired effects for the default-terrain forcing-product/feedback matrix, averaged over the five hourly spatial medians. Positive temperature values indicate a warmer model-minus-observation bias, positive endpoint-difference values indicate a larger 50–500 m potential-temperature difference, and positive wind-vector values indicate a larger layer-mean wind-vector magnitude. Values are descriptive means for this case.

Effect	ΔT2 Bias (°C)	$\Delta S_{50:500}$ (K)	$\Delta U_{50:500}$ (m s^{-1})
GFS − NAM, two-way	+1.79	−3.63	+0.79
GFS − NAM, one-way	+1.76	−3.65	+0.99
GFS one-way − two-way	+0.07	−0.07	+0.10
NAM one-way − two-way	−0.01	+0.00	+0.01
Feedback interaction	+0.04	−0.10	+0.10

Table A2. Primary-layer wind-metric comparison for the feedback-matched GFS-minus-NAM pairs. Each value is the mean of five hourly paired-station medians. The alignment ratio is the magnitude of the layer-mean vector divided by layer-mean scalar speed.

Feedback	Δ Vector Magnitude (m s^{-1})	Δ Scalar Speed (m s^{-1})	Δ Alignment Ratio
Two-way	+0.79	+0.78	+0.063
One-way	+0.99	+1.01	+0.053

Table A3. Layer-bound sensitivity of the GFS-minus-NAM contrasts at the primary 25 m wind integration increment. Each value is the mean of five hourly paired-station medians on d02 over the same eleven columns. ΔS is the potential-temperature endpoint-difference contrast; ΔU and ΔV are the layer-mean vector-magnitude and scalar-speed contrasts, respectively.

	ΔS (K)		ΔU (m s^{-1})		ΔV (m s^{-1})	
Layer (m AGL)	**Two-Way**	**One-Way**	**Two-Way**	**One-Way**	**Two-Way**	**One-Way**
50–500	−3.63	−3.65	+0.79	+0.99	+0.78	+1.01
50–250	−2.22	−2.08	+0.67	+0.85	+0.70	+0.83
100–500	−3.38	−3.42	+0.77	+1.02	+0.79	+1.02

Table A4. Surface, spatial, and vertical corroboration for the two GFS-minus-NAM forcing-product contrasts. All values use the 1400–1800 UTC window described in Section 3.5. The first four rows use station samples; the last four use area-weighted diagnostics over the basin-floor delineated area.

Diagnostic	**Result**	**Scope**
2 m temperature	GFS-minus-NAM bias contrast +1.79 °C two-way and +1.76 °C one-way; absolute-error differences +1.41 and +1.50 °C	Same ordering as the common-AGL endpoint difference
10 m wind	Bias contrast +0.91 and +0.96 m s^{-1}; hourly absolute-error differences change sign	No stable wind-skill ordering
Surface pressure	Absolute-error contrast −13.6 Pa two-way and −10.9 Pa one-way; negative in every hour	Opposite ordering from temperature, with small absolute differences
Two-site surface index	Observed 12.63–20.21 K; NAM 16.44–18.07 K; GFS 7.35–11.74 K under two-way feedback	Horizontally separated surface contrast, not inversion strength or depth
Mean 2 m potential-temperature difference	+0.89 to +1.16 K at 1400–1500 UTC; +2.00 to +2.74 K at 1600–1800 UTC	Positive at all ten pair-hours
Footprint sign share	0.64–0.67 at 1400–1500 UTC; 0.84–0.93 later	Early response is not spatially uniform
Low-to-aloft absolute-response ratio	1.45–3.12	Response concentrated in the lower atmosphere
50–500 m endpoint-difference contrast	−4.33 to −3.04 K	Negative at all ten pair-hours

Table A5. Footprint-mean, vertically integrated heat-deficit contrasts over the 1400–1800 UTC decision window. The primary definition uses the area convex hull with a 2200 m crest (4965 cells, 5277 km^2); envelopes span all footprint definitions from sensitivity tests. The values are deterministic descriptive contrasts in MJ m^{-2}.

GFS − NAM Quantity	**Two-Way Pair**	**One-Way Pair**
Mean difference (primary)	−1.23	−1.25
Hourly range (primary)	−1.50 to −0.97	−1.53 to −0.99
Hourly envelope (seven definitions)	−1.77 to −0.70	−1.80 to −0.71

Table A5. *Cont.*

GFS − NAM Quantity	**Two-Way Pair**	**One-Way Pair**
GFS change, 1400–1800 UTC	+0.32	+0.30
NAM change, 1400–1800 UTC	+0.15	+0.13
Difference in change (primary)	+0.17	+0.17

Table A6. Case-window temperature stations and nearest d02 sampling audit. Coordinates and reported elevations are from the registered Synoptic Data PBC metadata; separation is great-circle distance to the sampled cell centre, and the final column is X3 default-terrain model height minus reported station elevation.

ID	Site	Latitude	Longitude	Elevation (m)	Cell Distance (m)	Elevation Offset (m)
KVEL	Vernal Airport	40.44295	−109.51273	1604	405	0.0
QRS	Roosevelt	40.29430	−110.00900	1588	354	−32.4
QV4	Vernal	40.46472	−109.56083	1666	408	−2.0
USU01	Seven Sisters	39.98130	−109.34540	1615	555	−3.0
USU02	Gusher	40.29350	−109.65750	1550	230	−5.0
USU03	Mountain Home	40.43190	−110.38210	2228	594	−5.8
USU04	Seep Ridge	39.75390	−109.54600	1933	133	+2.9
USU05	Wells Draw	40.06700	−110.15100	1759	187	+10.1
USU07	Pariette Draw	40.03460	−109.83010	1465	485	−0.1
USU08	USU Research Trailer	40.14370	−109.46718	1570	446	−4.2
UUT01	Uintah Basin Research Trailer	40.26560	−110.04120	1561	419	+14.3

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